

HDOC DAY-11 ACTIVITY REPORT

Menu-Driven C Program Using Switch-Case

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Prepared as per the HDOC Day-11 instructions.

1. Problem Statement

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

1. Find the sum of squares of its digits using a **while loop**.
2. Find the sum of cubes of its digits using a **for loop**.
3. Find the difference between the sum of cubes and the sum of squares of its digits using a **do-while loop**.

Input

A positive integer **num** and a menu choice **1, 2, or 3**.

Output

Depending on the selected choice, display the sum of squares, sum of cubes, or the difference between the two sums.

Example for num = 123

Sum of squares = 14

Sum of cubes = 36

Difference = 22

2. Algorithm

1. Start.
2. Declare integer variables: num, temp, digit, choice, sumSquare = 0, sumCube = 0, digits = 0.
3. Read a positive integer num.
4. Display the menu and read choice.
5. Use switch(choice).
6. For choice 1: use a while loop to extract each digit and add its square to sumSquare; display the result.
7. For choice 2: count digits, then use a for loop to extract each digit and add its cube to sumCube; display the result.
8. For choice 3: use a do-while loop to calculate sumSquare and sumCube; display sumCube - sumSquare.
9. For any other choice, display Invalid Choice.
10. Stop.

3. Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	num = 123 choice = 1	Sum of squares = 14	Sum of squares = 14	Pass
2	num = 123 choice = 2	Sum of cubes = 36	Sum of cubes = 36	Pass

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
3	num = 123 choice = 3	Difference = 22	Difference = 22	Pass
4	num = 45 choice = 1	Sum of squares = 41	Sum of squares = 41	Pass
5	num = 45 choice = 2	Sum of cubes = 189	Sum of cubes = 189	Pass
6	num = 45 choice = 3	Difference = 148	Difference = 148	Pass

4. Program Code

The following screenshots show the C program written for the activity.



```
UW PICO 5.09 File: day11.cpp
#include <stdio.h>

int main() {
    int num, temp, digit, choice;
    int sumSquare = 0, sumCube = 0, digits = 0;
    printf("Enter a positive integer: ");
    scanf("%d", &num);
    printf("\nMENU\n");
    printf("1. Sum of squares of digits\n");
    printf("2. Sum of cubes of digits\n");
    printf("3. Difference (Cubes - Squares)\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    switch(choice) {
        case 1:
            temp = num;
            while(temp > 0) {
                digit = temp % 10;
                sumSquare += digit * digit;
                temp /= 10;
            }
            printf("Sum of squares = %d\n", sumSquare);
            break;
        case 2:
            temp = num;
            while(temp > 0) {
                digits++;
                temp /= 10;
            }
    }
}
```

Get Help Exit WriteOut Justify Read File Where is Prev Pg Next Pg Out Text UnOut Text Cur Pos To Spell

UW PICO 5.09File: day11.cpp

```
temp = num;

while(temp > 0) {
    digit = temp % 10;
    sumSquare += digit * digit;
    temp /= 10;
}

printf("Sum of squares = %d\n", sumSquare);
break;

case 2:
temp = num;
while(temp > 0) {
    digits++;
    temp /= 10;
}

temp = num;
for(int i = 1; i <= digits; i++) {
    digit = temp % 10;
    sumCube += digit * digit * digit;
    temp /= 10;
}

printf("Sum of cubes = %d\n", sumCube);
break;

case 3:
temp = num;
do {
    digit = temp % 10;
    sumSquare += digit * digit;
    sumCube += digit * digit * digit;
```

Get Help

Exit

WriteOut

Justify

Read File

Where is

Prev Pg

Next Pg

Cut Text

UnCut Text

Cur Pos

To Spell

```
UW PICO 5.09 File: day11.cpp

temp = num;
for(int i = 1; i <= digits; i++) {
    digit = temp % 10;
    sumCube += digit * digit * digit;
    temp /= 10;
}
printf("Sum of cubes = %d\n", sumCube);
break;

case 3:
temp = num;
do {
    digit = temp % 10;
    sumSquare += digit * digit;
    sumCube += digit * digit * digit;
    temp /= 10;
} while(temp > 0);
printf("Difference = %d\n", sumCube - sumSquare);
break;

default:
    printf("Invalid Choice!\n");
}
return 0;
}
```

Get Help
Exit

WriteOut
Justify

Read File
Where is

Prev Pg
Next Pg

Cut Text
UnCut Text

Cur Pos
To Spell

5. Compilation, Execution and Testing

The program was compiled and executed successfully. The execution screenshot verifies all three required operations using the input 123.

```
bhavya@Bhavyas-MacBook-Air ~ % nano day11.cpp
bhavya@Bhavyas-MacBook-Air ~ % clang day11.cpp -o day11
bhavya@Bhavyas-MacBook-Air ~ % ./day11
Enter a positive integer: 123

MENU
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference (Cubes - Squares)
Enter your choice: 1
Sum of squares = 14
bhavya@Bhavyas-MacBook-Air ~ % ./day11
Enter a positive integer: 123

MENU
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference (Cubes - Squares)
Enter your choice: 2
Sum of cubes = 36
bhavya@Bhavyas-MacBook-Air ~ % ./day11
Enter a positive integer: 123

MENU
1. Sum of squares of digits
2. Sum of cubes of digits
3. Difference (Cubes - Squares)
Enter your choice: 3
Difference = 22
bhavya@Bhavyas-MacBook-Air ~ %
```

Observed results: Choice 1 gives 14, Choice 2 gives 36, and Choice 3 gives 22. These match the expected results, so the test cases pass.